Manan Agrawal

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SUMMARY

AI Engineer and Machine Learning Engineer focused on building production grade AI systems, Generative AI applications, and scalable ML pipelines. Experience with LLM orchestration, agentic workflows, retrieval augmented generation, and deploying end to end AI services with strong emphasis on performance, reliability, and evaluation.

WORK EXPERIENCE

Research and Development AI Intern, Martinrea International US Inc. *2025 – 2026*

- Led end to end AI solution lifecycle by identifying integration opportunities in enterprise software, proposing AI use cases, and deploying production grade ML systems for anomaly detection and forecasting.
- Built scalable ML pipelines covering ingestion, feature engineering, inference, evaluation, and monitoring, processing 500K+ records daily.
- Developed and integrated distributed data pipelines and backend AI services with APIs, databases, and enterprise systems enabling real time workflows.
- Collaborated with product and business stakeholders to translate requirements into scalable AI solutions integrated into existing systems.
- Conducted experimentation, model validation, and performance analysis to ensure reliability and robustness of deployed AI systems.
- Improved system performance by 35% and reduced latency by 25% through optimization and efficient system design.

GenAI Project Intern, Honeywell *01/2025 – 05/2025*

- Designed and deployed production grade AI system for real time email classification using LLMs, processing 1000+ requests/day with 98% + accuracy and reducing manual effort by 60%.
- Built RAG pipeline using embeddings and vector similarity (top K retrieval) to ground predictions with historical labeled data.
- Developed hierarchical classification system improving accuracy, precision, and recall across enterprise workflows.
- Implemented confidence scoring with human in the loop validation using thresholding and calibration.
- Designed evaluation and monitoring pipelines to track model performance, optimize thresholds, and enable continuous improvement.
- Built scalable pipelines integrating APIs, vector databases, and enterprise systems for real time inference and monitoring.

TECHNICAL SKILLS

- **Languages:** Python, Java, C++, SQL, JavaScript, TypeScript
- **Core Concepts:** Machine Learning, Deep Learning, NLP, Transformers, Statistical Modeling, Causal Inference, A/B Testing
- **Frameworks:** PyTorch, TensorFlow, FastAPI, MLflow, LangChain, LlamaIndex, AutoGen
- **AI Systems:** Generative AI, LLMs, Agentic AI, Multi Agent Systems, Retrieval Augmented Generation
- **Data Systems:** SQL, NoSQL, Data Pipelines, Feature Engineering, Large Scale Processing
- **Vector Databases:** FAISS, Pinecone, Weaviate
- **MLOps:** CI CD, Model Monitoring, Prompt Lifecycle, Experiment Tracking, Versioning
- **Systems:** Distributed Systems, REST APIs, Async Programming, Scalable Systems
- **Cloud:** AWS, GCP (familiar), Docker, Kubernetes

PROJECTS

LLM Based Document Understanding and Retrieval System

- Built production ready retrieval system using transformer embeddings and FAISS enabling sub second semantic search.
- Designed pipelines for extracting structured insights from unstructured data using LLMs and vector search.

AI Powered Trading and Decision System

- Developed time series forecasting and probabilistic models for financial decision systems.
- Designed automated decision engine using prediction confidence, risk modeling, and expected value estimation with backtesting.

HONORS AND PUBLICATIONS

- IEEE Peer Reviewer for machine learning and AI conferences and journals
- Published 15+ research papers in machine learning, AI, and applied statistics
- Govt. of India Design Patent No 386153 001
- Winner, Most Complete Solution, LPL Financial Hackathon (Alpha: AI platform)
- Invited Guest Speaker, Johnson C. Smith University, Charlotte (AI and Finance)

EDUCATION

MS Computer Engineering, UNC Charlotte *08/2024 – 05/2026*
BTech Electronics Engineering, RCOEM, Nagpur University *07/2020 – 06/2024*